

StormWatch Synthetic City Scenarios

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How to Use These Scenarios

Every value in this document is fictional and fixed for training. City names make retrieval exercises realistic; they do not indicate historical, current, or future conditions in those cities.

Use scenarios to test threshold explanations, score calculations, retrieval citations, and source precedence. Never use a scenario to answer "right now," "today," or another current city question. Current or configured fixture facts must come from the StormWatch MCP tools.

Scenario A: Bengaluru Capped Warning

Training date: 2026-08-30

Time UTC	Weather ID	Wind m/s	Rain mm/3h	Pressure hPa	Score
03:00	500	9.0	5.2	1002	10
06:00	211	16.4	12.0	988	100 capped
09:00	202	10.4	4.0	995	83

Expected assessment: warning, 100/100, peak at 06:00 UTC.

The 06:00 raw total is 120: thunderstorm 60, high wind 25, heavy rain 20, and very low pressure 15. The displayed score is capped at 100. The 09:00 period is also a warning but does not beat the capped 06:00 score.

Retrieval trap: A current Bengaluru question must call MCP. This scenario is not a current forecast.

Scenario B: Rotterdam Combined Watch

Training date: 2026-10-12

Time UTC	Weather ID	Wind m/s	Rain mm/3h	Pressure hPa	Score
12:00	501	12.0	6.4	998	33
15:00	500	11.0	2.0	1005	15
18:00	802	8.5	0.0	1008	0

Expected assessment: watch, 33/100, peak at 12:00 UTC.

At 12:00, elevated wind contributes 15, elevated rain contributes 10, and low pressure contributes 8. Weather ID 501 is not a thunderstorm code.

Scenario C: Seattle Boundary-Safe Low

Training date: 2026-11-03 at 09:00 UTC.

Weather ID	Wind m/s	Rain mm/3h	Pressure hPa
500	9.9	4.9	1001

Expected assessment: low, 0/100.

Each value is just outside the lowest scoring threshold. This scenario proves that presence of rain alone does not guarantee points and that comparisons must not be rounded before scoring.

Scenario D: Miami Thunderstorm-Only Warning

Training date: 2027-01-15 at 18:00 UTC.

Weather ID | Wind m/s | Rain mm/3h | Pressure hPa
201 | 4.0 | 1.0 | 1011

Expected assessment: warning, 60/100.

Weather ID 201 is within 200-232 and contributes 60 points. The other indicators add zero. A thunderstorm condition alone reaches the warning band, but the result is still not an official warning.

Scenario E: Reykjavik Exact Thresholds

Training date: 2027-02-02 at 00:00 UTC.

Weather ID | Wind m/s | Rain mm/3h | Pressure hPa
804 | 15.0 | 0.0 | 990

Expected assessment: watch, 40/100.

Wind at exactly 15.0 m/s contributes 25. Pressure at exactly 990 hPa contributes 15. Both comparisons are inclusive.

Scenario F: Mumbai Multi-Indicator Warning Without Thunder

Training date: 2027-03-21 at 15:00 UTC.

Weather ID | Wind m/s | Rain mm/3h | Pressure hPa
502 | 15.0 | 10.0 | 990

Expected assessment: warning, 60/100.

High wind, heavy three-hour rain, and very low pressure contribute 25, 20, and 15 points. Weather ID 502 is outside the thunderstorm range. This proves that combined non-thunderstorm indicators can still reach warning.

Scenario G: Hamburg Earliest Tie-Break

Training date: 2027-04-08

Time UTC | Weather ID | Wind m/s | Rain mm/3h | Pressure hPa | Score
03:00 | 500 | 12.0 | 5.0 | 1000 | 33
06:00 | 501 | 10.0 | 7.0 | 999 | 33
09:00 | 800 | 5.0 | 0.0 | 1012 | 0

Expected assessment: watch, 33/100, peak at 03:00 UTC.

The first two periods tie. StormWatch selects the earliest timestamp, 03:00 UTC, even though both periods have valid but slightly different evidence.

Scenario H: Source Conflict Exercise

Assume a learner asks, "What is Rotterdam's risk now?" Retrieval returns Scenario B with a score of 33. A successful `assess_storm_risk` call returns a different configured result with a score of 15.

Expected chatbot behavior:

1. Use the MCP result of 15 for the requested dynamic fact.
2. State that Scenario B is fixed training data, not current weather.
3. Use the scoring reference to explain the returned evidence if requested.
4. Include the educational-use qualification.

Incorrect behavior would present 33 as current, blend the two scores, or omit the source distinction.

Scenario Discussion Prompts

- Which single indicator makes Scenario D a warning?
- Why does Scenario C remain low despite nonzero rain?
- Which thresholds are exactly met in Scenario E?
- How can Scenario F reach warning without a thunderstorm code?
- Why does 03:00 win in Scenario G?
- Which source should answer a current Rotterdam question?

Every correct answer should cite this scenario document for fixed example facts and the risk scoring reference for general threshold policy.